

Signals, Convolution & Z-Transforms

Revision notes with 10 fully solved questions

Scope: continuous-time signals, sampling and Nyquist rate, discrete-time indexing, discrete- and continuous-time convolution, zero-padding and introductory Z-transforms. Includes the material from *IDSA Assignment 3 pt1*, with 10 main solved questions (some multipart) across Sections 1–7.

Conventions: finite sequences are zero outside their listed indices. Unlabelled lists start at $n = 0$. In a stem plot, horizontal position is the sample index and vertical height is the sample value. Here $u[0] = 1$.

1 Read a signal before calculating

Continuous time: what does $x(t)$ mean?

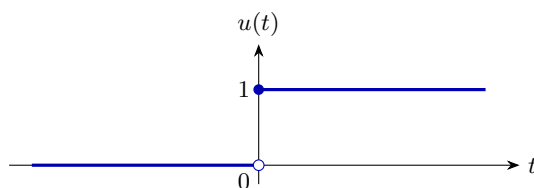
A **signal** describes how a quantity changes. For example, $x(t)$ could be a voltage measured at time t . A **continuous-time (CT) signal** has a value for every real time in its domain: $t = 0, 0.1, 0.12$, and every time between them. In a graph, the horizontal axis is time t and the vertical axis is the signal value (its **amplitude**).

Continuous time does not mean a smooth graph. A CT signal can jump. “Continuous-time” describes the time variable, not whether $x(t)$ is a continuous function. We use parentheses, $x(t)$, for CT signals and square brackets, $x[n]$, for discrete-time signals.

1. Unit step: a switch that turns on at zero.

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases} \quad u(t-a) \text{ turns on at } t = a.$$

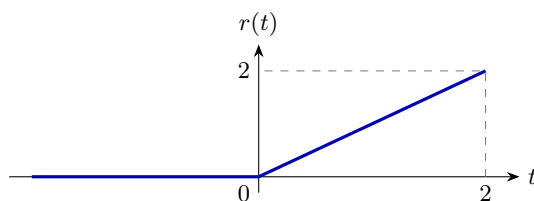
We choose $u(0) = 1$ in these notes. A filled dot marks an included endpoint; an open circle marks an excluded endpoint. No vertical connecting line is part of the step's graph.



2. Unit ramp: zero before the origin, then a straight rise.

$$r(t) = t u(t) = \begin{cases} 0, & t < 0, \\ t, & t \geq 0. \end{cases}$$

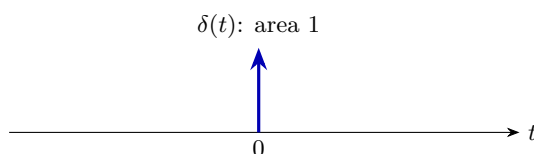
Its slope is 1: increasing time by one unit increases the amplitude by one unit.



3. Unit impulse: an ideal event with unit area. The CT impulse $\delta(t)$ is an idealized *distribution*, not an ordinary function with a finite peak. It is zero away from $t = 0$ and satisfies

$$\int_{-\infty}^{\infty} \delta(t) dt = 1, \quad \int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)$$

for f continuous at a . The arrow below is a symbol; its label gives **area (weight)**, not height. In contrast, $\delta[n]$ is an ordinary discrete sequence with value 1 at $n = 0$.

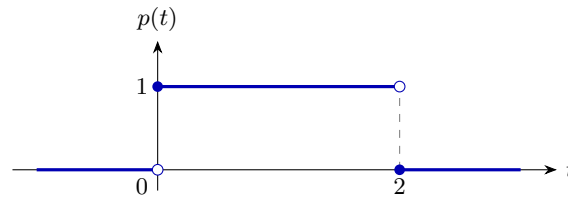


More continuous-time signals and their graphs

4. Rectangular pulse: switch on, then switch off. A pulse of height A from $t = a$ to $t = b$ ($a < b$) is

$$p(t) = A[u(t - a) - u(t - b)] = \begin{cases} A, & a \leq t < b, \\ 0, & \text{otherwise.} \end{cases}$$

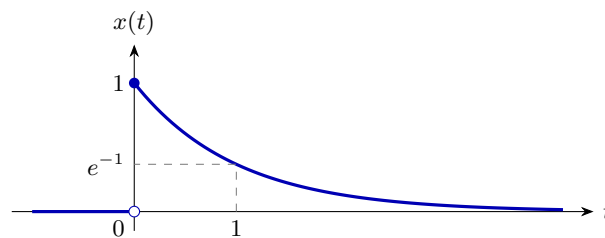
The first step turns it on; subtracting the second turns it off. Its width is $b - a$ and its area is $A(b - a)$. Below, $A = 1$, $a = 0$, $b = 2$.



5. Decaying exponential: a value that gradually dies away.

$$x(t) = e^{-at}u(t), \quad a > 0.$$

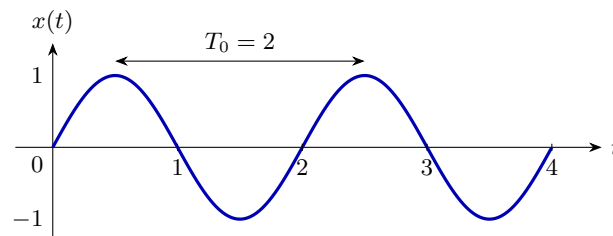
It is zero for $t < 0$, starts at 1 at $t = 0$, and approaches zero as $t \rightarrow \infty$. Larger a means faster decay. The **time constant** is $\tau = 1/a$; at $t = \tau$, the value is $e^{-1} \approx 0.368$. The graph uses $a = 1$.



6. Sinusoid: a repeating oscillation.

$$x(t) = A \sin(\omega_0 t + \phi), \quad \omega_0 = 2\pi f_0, \quad T_0 = \frac{1}{f_0} = \frac{2\pi}{\omega_0}.$$

For $A > 0$, A is the peak amplitude. With time measured in seconds, $f_0 > 0$ is frequency in hertz (cycles per second), ω_0 is angular frequency in radians per second, and ϕ is phase in radians (the starting position in the cycle). The **period** T_0 is the time for one complete cycle. The graph is $x(t) = \sin(\pi t)$, so $A = 1$, $\phi = 0$, $f_0 = 1/2$ and $T_0 = 2$.



A signal is **periodic** if $x(t + T) = x(t)$ for every t , for some $T > 0$. A sinusoid repeats for all time; only two cycles are displayed above. A single rectangular pulse and a decaying exponential are aperiodic.

How to shift, reflect and scale a continuous-time graph

Change the **outside** of $x(\cdot)$ to change heights. Change its **argument** to move points along the time axis. For $t_0 > 0$ and $a > 0$:

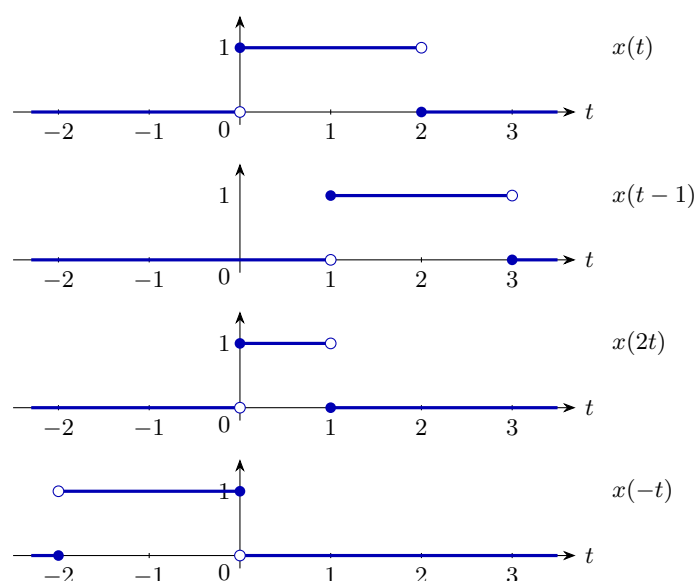
Expression	Effect on the original graph
$Cx(t)$	Multiply every height by C ; negative C also reflects vertically.
$x(t - t_0)$	Delay: move the graph right by t_0 .
$x(t + t_0)$	Advance: move the graph left by t_0 .
$x(-t)$	Time reversal: reflect across the vertical axis.
$x(at)$	Divide every time coordinate by a : compress if $a > 1$, stretch if $0 < a < 1$.

Reliable rule for a combined change: for $y(t) = Cx(at + b)$, with $a \neq 0$, a point originally at time s moves to

$$t = \frac{s - b}{a}, \quad \text{new height} = Cx(s).$$

This comes directly from setting the argument $at + b = s$. If $a < 0$, the order of the endpoints reverses. Apply this to the start, end, peak and every corner of a graph.

Graph illustration. Let $x(t) = u(t) - u(t - 2)$, the unit-height pulse on $0 \leq t < 2$. Each panel uses the same time scale; endpoint dots follow the chosen $u(0) = 1$ convention.



The nonzero intervals are $[0, 2)$, $[1, 3)$, $[0, 1)$ and $(-2, 0]$, respectively. Here $[]$ includes an endpoint and $()$ excludes it. Time scaling changes the width, not the height.

Connecting CT to discrete time. Sampling a CT signal every $T_s > 0$ time units gives $x[n] = x(nT_s)$, where n is an integer. A CT graph shows values at all times; a discrete-time stem plot shows values only at the allowed indices. Joining stems with a line does not define the missing values.

Sampling: converting a continuous-time signal into a sequence

Sampling means reading a signal at equally spaced times. If the spacing is T_s seconds, the sampling times are $\dots, -T_s, 0, T_s, 2T_s, \dots$ and

$$x[n] = x(t)|_{t=nT_s} = x(nT_s), \quad T_s = \frac{1}{f_s}.$$

Here n is an integer sample index, T_s is the **sampling interval** (seconds per sample), and f_s is the **sampling rate** (samples per second, conventionally expressed in Hz). Sampling chooses times; it does not round the amplitudes. Rounding amplitudes is a separate operation called quantization.

Three frequencies that must not be confused:

Symbol	Units	Meaning and conversion
f	Hz	Cycles per second in the original CT signal.
ω	rad/s	CT angular frequency: $\omega = 2\pi f$.
Ω	rad/sample	DT angular frequency: $\Omega = \omega T_s = 2\pi f / f_s$.

Thus the general sampling rule for a sinusoid is

$$A \cos(2\pi f t + \phi) \longrightarrow A \cos\left(2\pi \frac{f}{f_s} n + \phi\right).$$

The amplitude A and phase ϕ stay the same; replace *every* t by n/f_s .

Nyquist rate and why the highest frequency matters

A **band-limited** signal has no frequency components above some highest frequency f_m . Ideal uniform samples allow exact reconstruction when the sampling rate is greater than twice that highest frequency, with ideal reconstruction and no noise:

$$\text{Nyquist rate} = 2f_m, \quad f_s > 2f_m \text{ avoids the boundary case.}$$

Your assignment writes the threshold rule as $f_s \geq 2f_m$ and uses $f_s = 2f_m$ in its examples. We use those same rates below to reproduce the assignment's sequences. **Equality needs care:** a sinusoid exactly at $f_s/2$ can lose phase information. For example, $\sin(2\pi f_m n / (2f_m)) = \sin(\pi n) = 0$ at every sample. So equality is not a universal guarantee of perfect reconstruction.

The **Nyquist frequency** of a chosen sampler is $f_s/2$. This is different from the **Nyquist rate** $2f_m$ of a signal. **Aliasing** means different CT frequencies produce the same samples; undersampling can make a fast oscillation appear slower. An anti-aliasing low-pass filter removes unwanted high frequencies before sampling.

For a sum of sinusoids, find each $f = |\omega|/(2\pi)$ and take the largest. Amplitudes do not determine the Nyquist rate. A constant term is **DC**, meaning frequency zero.

1. Read the coefficient of t inside each sine or cosine; that is ω .
2. Divide by 2π to get Hz; choose the largest f_m .
3. Compute $2f_m$. Use the specified f_s , or state the rate you choose.
4. Put $T_s = 1/f_s$ and substitute $t = nT_s$ in the entire expression.
5. Use radians to calculate samples; plot each value at its integer index.

If only a Nyquist rate is requested, report $2f_m$ in Hz. If $x[n]$ is also requested but f_s is unspecified, its formula depends on the rate chosen; state that choice explicitly.

Question 1 — Sampling a single cosine: Assignment 3, sampling Q1

Given $x(t) = 2 \cos(10\pi t)$, find the Nyquist rate and the sampled sequence. Use the assignment's choice $f_s = 2f_m$ and plot the result.

Solution. Compare with $A \cos(2\pi ft + \phi)$:

$$A = 2, \quad \phi = 0, \quad 2\pi f_m = 10\pi \implies f_m = 5 \text{ Hz.}$$

The amplitude 2 does not affect this frequency. The CT period is $1/5 = 0.2$ s.

Nyquist rate = $2(5) = 10$ Hz.

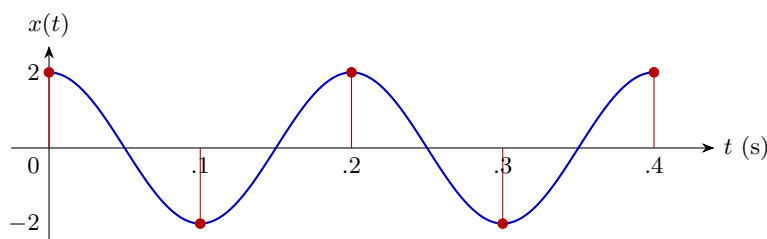
Choose $f_s = 10$ Hz as in the assignment. Then $T_s = 1/10 = 0.1$ s and

$$\begin{aligned} x[n] &= x(nT_s) = 2 \cos(10\pi nT_s) \\ &= 2 \cos\left(10\pi n \frac{1}{10}\right) = \boxed{2 \cos(\pi n) = 2(-1)^n}. \end{aligned}$$

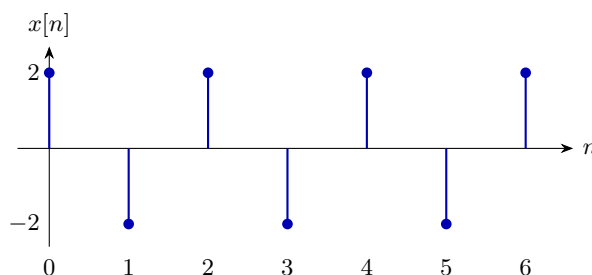
Even indices give 2 and odd indices give -2 :

n	Time nT_s (s)	Calculation	$x[n]$
0	0	$2 \cos 0$	2
1	0.1	$2 \cos \pi$	-2
2	0.2	$2 \cos 2\pi$	2
3	0.3	$2 \cos 3\pi$	-2
4	0.4	$2 \cos 4\pi$	2

CT curve and the sampling instants



Discrete sequence: $x[n] = 2(-1)^n$



Check: $x[n+2] = x[n]$, so the sequence has period 2 samples. It exists for all integer n ; only a short segment is drawn. At any other rate, use $x[n] = 2 \cos(10\pi n/f_s)$ instead.

Question 2 — Sampling a sum of cosines: Assignment 3, sampling Q2

Given

$$x(t) = 5 + 3 \cos(100\pi t) + 7 \cos(1200\pi t),$$

find the Nyquist rate and the sampled sequence, calculate sample values, and draw its stem plot using the assignment's threshold rate.

Solution. The constant 5 has frequency zero. Divide each angular frequency by 2π :

$$f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz}, \quad f_2 = \frac{1200\pi}{2\pi} = 600 \text{ Hz}.$$

Therefore $f_m = \max(0, 50, 600) = 600 \text{ Hz}$ and

$$\boxed{\text{Nyquist rate} = 2(600) = 1200 \text{ Hz.}}$$

Choose $f_s = 1200 \text{ Hz}$, giving $T_s = 1/1200 \text{ s} \approx 0.8333 \text{ ms}$. Substitute $t = n/1200$ into both cosine terms:

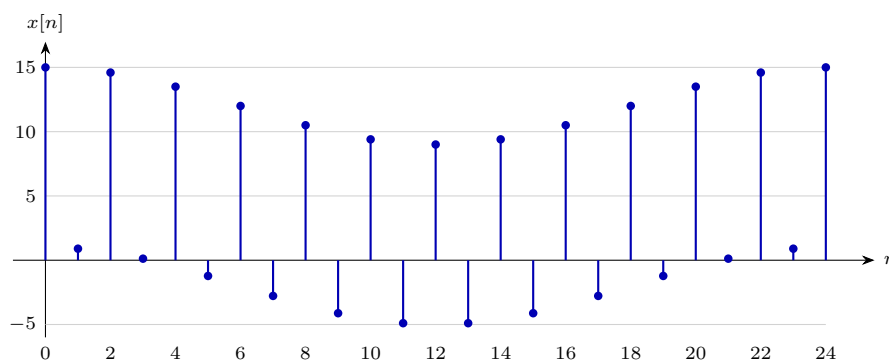
$$\begin{aligned} x[n] &= 5 + 3 \cos\left(\frac{100\pi n}{1200}\right) + 7 \cos\left(\frac{1200\pi n}{1200}\right) \\ &= \boxed{5 + 3 \cos\left(\frac{\pi n}{12}\right) + 7 \cos(\pi n)} \\ &= 5 + 3 \cos\left(\frac{\pi n}{12}\right) + 7(-1)^n. \end{aligned}$$

The two DT angular frequencies are $\pi/12$ and π radians/sample. The DC value remains 5.

n	Calculation	$x[n]$ (approximately)
-1	$5 + 3 \cos(-\pi/12) - 7$	0.89778
0	$5 + 3(1) + 7$	15
1	$5 + 3 \cos(\pi/12) - 7$	0.89778
2	$5 + 3 \cos(\pi/6) + 7$	14.59808
3	$5 + 3 \cos(\pi/4) - 7$	0.12132
4	$5 + 3 \cos(\pi/3) + 7$	13.5

For instance, $x[1] = 5 + 3(0.9659258) - 7 = 0.8977774$. Use radian mode when entering $\pi/12$ into a calculator. Since cosine is even, $x[-n] = x[n]$.

One full period, with the repeated endpoint at $n = 24$



Check: $\cos(\pi n/12)$ repeats every 24 samples and $\cos(\pi n)$ every 2 samples. The combined sequence has period $\text{lcm}(24, 2) = 24$, so $x[24] = x[0] = 15$. As in Question 1, the rate used is exactly the textbook threshold; choose $f_s > 1200 \text{ Hz}$ to avoid its boundary ambiguity.

Discrete-time indexing

A **sample** is one measured or listed value. A **sequence** is an ordered collection of samples. The integer n labels the position, while $x[n]$ gives the value at that position: $x[2] = 5$ means “the sample at index 2 has value 5.” A negative index means a position before the chosen origin; it does not mean a negative amplitude. A zero sample is still a position and must not be deleted.

Reading the notation. In $\{a, b, c\}_{n=-1,0,1}$, the first entry a belongs to $n = -1$, the middle entry b to $n = 0$, and the last entry c to $n = 1$. The subscript is an index list, not another set of signal values. A stem is a vertical line from the horizontal axis to a sample’s value; negative values extend below the axis.

A discrete-time signal $x[n]$ assigns a value to each integer index n . An arrow under a sequence entry marks $n = 0$. Always write the index row before calculating.

- $\delta[n - r]$ is 1 at $n = r$ and zero elsewhere.
- $u[n]$ is 1 for $n \geq 0$ and zero for $n < 0$.
- $u[n] - u[n - L]$ is a length- L rectangle: 1 at $n = 0, \dots, L - 1$.
- For $v[n] = x[an + b]$, a sample originally at m appears where $an + b = m$, if that gives an integer n ; its height stays unchanged.

Question 3 — Indexing, shifting and reflection

Given $x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}$, find $x[-1]$, $x[3]$, $x[n+1]$ and $x[1-n]$. Also express $x[n]$ as shifted impulses.

Solution. The original samples are

$$\begin{array}{c|cccc} n & -2 & -1 & 0 & 1 \\ \hline x[n] & 1 & 2 & 2 & 1 \end{array}.$$

Thus $\boxed{x[-1] = 2, \quad x[3] = 0}$, since 3 lies outside the listed range.

For $v[n] = x[n+1]$, solve $n+1 = m$, giving $n = m-1$. For $w[n] = x[1-n]$, solve $1-n = m$, giving $n = 1-m$:

Original index m	Height $x[m]$	Index in $v[n]$	Index in $w[n]$
-2	1	-3	3
-1	2	-2	2
0	2	-1	1
1	1	0	0

Reading each result in increasing index order,

$$\boxed{x[n+1] = \{1, 2, 2, 1\}_{n=-3, -2, -1, 0}}, \quad \boxed{x[1-n] = \{1, 2, 2, 1\}_{n=0, 1, 2, 3}}.$$

Place an impulse at each original index and multiply it by that sample's height:

$$\boxed{x[n] = \delta[n+2] + 2\delta[n+1] + 2\delta[n] + \delta[n-1]}.$$

Check: substituting $n = -2, -1, 0, 1$ gives 1, 2, 2, 1, respectively. Every other index gives zero.

2 Linear convolution: flip, shift, multiply, add

A **system** is a rule or device that converts an input signal $x[n]$ into an output signal $y[n]$. It might represent a filter, an echo or a measurement device. The input is what we apply; the output is the resulting response.

A system is **linear** if adding inputs adds their outputs, and multiplying an input by a constant multiplies its output by the same constant. If inputs x_1, x_2 produce outputs y_1, y_2 , then input $ax_1 + bx_2$ produces output $ay_1 + by_2$. It is **time-invariant** if delaying an input simply delays its output by the same amount. **LTI** means both properties hold.

The **impulse response** $h[n]$ is the output produced by the unit impulse $\delta[n]$. The letter h does not mean a second time variable: it names a signal, just as x and y do. An input sample of height $x[k]$ at index k is $x[k]\delta[n-k]$. Time invariance shifts its response to $h[n-k]$; linearity scales that response by $x[k]$ and lets us add all such responses.

For a linear time-invariant system with impulse response $h[n]$, each input sample produces a scaled, shifted copy of $h[n]$. Adding these copies gives

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

The symbol $*$ denotes **convolution**. The capital sigma $\sum$ means “add the terms.” For each fixed n , run through all integer k , calculate $x[k]h[n-k]$, and add. The limits $-\infty$ and ∞ mean all integer indices; with finite inputs, almost all terms are zero. Ordinary pointwise multiplication would instead be $x[n]h[n]$.

Here n is the output index and k is the summation index. Convolution is a sum of overlap products, not pointwise multiplication.

1. **Index:** if the input ranges are $a, \dots, b$ and $c, \dots, d$, the output lies within $a + c, \dots, b + d$.
2. **Flip:** change $h[k]$ to $h[-k]$.
3. **Shift:** form $h[n-k]$; a sample of h originally at m is now at $k = n - m$.
4. **Multiply and add:** multiply at matching k and sum. Repeat for each n .

For listed lengths L, M , retain $L + M - 1$ output positions, including any zero entries. This follows by counting the integers from $a + c$ to $b + d$:

$$(b + d) - (a + c) + 1 = (b - a + 1) + (d - c + 1) - 1 = L + M - 1.$$

The **support** is the set of indices where a signal is nonzero. Listed arrays may also include zero entries, so their stored length and their nonzero support need not be identical.

Question 4 — Linear convolution with negative indices

Find $y = x * h$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1}.$$

Solution. The output range is $-2 + (-1) = -3$ to $1 + 1 = 2$, with $4 + 3 - 1 = 6$ positions. The flipped sequence $h[-k]$ has heights 2, 3, 1 at $k = -1, 0, 1$.

Using the four possible nonzero samples of x ,

$$y[n] = h[n+2] + 2h[n+1] + 2h[n] + h[n-1].$$

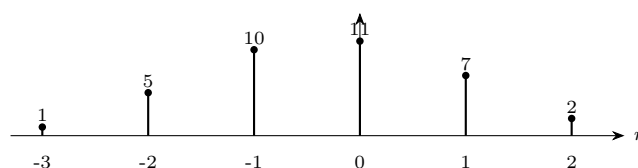
Substitute each output index; only nonzero products are shown:

n	Overlap products	$y[n]$
-3	1(1)	1
-2	1(3) + 2(1)	5
-1	1(2) + 2(3) + 2(1)	10
0	2(2) + 2(3) + 1(1)	11
1	2(2) + 1(3)	7
2	1(2)	2

For instance, at $n = -2$, the shifted sequence $h[-2 - k]$ has heights 2, 3, 1 at $k = -3, -2, -1$. Only $k = -2, -1$ overlap x , so $y[-2] = 1(3) + 2(1) = 5$.

$$y = \{1, 5, 10, 11, 7, 2\}_{n=-3, -2, -1, 0, 1, 2}.$$

Output $y[n]$



Check: $\sum_n y[n] = 1 + 5 + 10 + 11 + 7 + 2 = 36 = (1 + 2 + 2 + 1)(1 + 3 + 2)$. Outside $[-3, 2]$, there is no overlap and $y[n] = 0$.

3 Overlap counting and shifted rows

Overlap means positions where both $x[k]$ and the shifted, reversed signal $h[n-k]$ are nonzero. A zero in either factor makes their product zero. Each output is the sum of the products in that overlap, including any negative products.

For two rectangles of unit height, every nonzero product is 1, so convolution counts overlapping samples. For general sequences, a shifted-row table organizes the same multiply-add calculation.

Question 5 — Overlap counting and signed shifted rows

(a) Convolution of two rectangular sequences. Find $y = x * h$ when $x = h = \{1, 1, 1\}$, both starting at $n = 0$.

Solution. Both inputs occupy 0, 1, 2, so y occupies 0, ..., 4. The fixed sequence $x[k]$ is nonzero for $0 \leq k \leq 2$. The moving sequence $h[n-k]$ is nonzero when $0 \leq n-k \leq 2$, or $n-2 \leq k \leq n$.

n	Nonzero k for $h[n-k]$	Overlapping k	Sum	$y[n]$
0	-2, -1, 0	{0}	1	1
1	-1, 0, 1	{0, 1}	1 + 1	2
2	0, 1, 2	{0, 1, 2}	1 + 1 + 1	3
3	1, 2, 3	{1, 2}	1 + 1	2
4	2, 3, 4	{2}	1	1

$$y = \{1, 2, 3, 2, 1\}_{n=0,1,2,3,4}.$$

Check: there are $3 + 3 - 1 = 5$ output positions and $1 + 2 + 3 + 2 + 1 = 9 = 3 \cdot 3$.

(b) Signed samples using shifted rows. Find $\{2, -1, 1\} * \{1, 2\}$, with both lists starting at zero.

Solution. The output has $3 + 2 - 1 = 4$ positions. Since $x[0] = 2$, $x[1] = -1$ and $x[2] = 1$,

$$y[n] = 2h[n] - h[n-1] + h[n-2].$$

Write each scaled copy of h in its correct columns, then add:

Output index n	0	1	2	3
$2h[n]$	2	4	0	0
$-h[n-1]$	0	-1	-2	0
$h[n-2]$	0	0	1	2
$y[n]$	2	3	-1	2

Thus $y[0] = 2$, $y[1] = 4 - 1 = 3$, $y[2] = -2 + 1 = -1$ and $y[3] = 2$, giving

$$y = \{2, 3, -1, 2\}_{n=0,1,2,3}.$$

Check: $2 + 3 - 1 + 2 = 6 = (2 - 1 + 1)(1 + 2)$.

Useful properties and checks

$$x * h = h * x, \quad x * (h_1 + h_2) = x * h_1 + x * h_2, \quad x * \delta[n-r] = x[n-r].$$

Check the output length, the first and last products, and $\sum y = (\sum x)(\sum h)$. A correct total alone does not verify the order of the samples. Keep interior zeros and track the origin explicitly.

4 Circular convolution: rotating circles

Circular means that the index positions behave like a clock. There are N stored positions, numbered $0, 1, \dots, N-1$. Moving one position after $N-1$ returns to 0; moving one position before 0 gives $N-1$. The word “point” means one sample position, so a 4-point result contains four samples.

The notation $a \bmod N$ means the remainder in $\{0, \dots, N-1\}$ obtained by adding or subtracting whole multiples of N . For example, $-1 + 4 = 3$, so $(-1) \bmod 4 = 3$; $5 - 4 = 1$, so $5 \bmod 4 = 1$.

An N -point circular convolution wraps all indices modulo N :

$$y_N[n] = \sum_{k=0}^{N-1} x[k]h[(n-k) \bmod N], \quad 0 \leq n < N.$$

Modulo N returns an index in $0, \dots, N-1$; for example, $(-1) \bmod 4 = 3$. The stored answer contains N samples; its periodic extension repeats these samples.

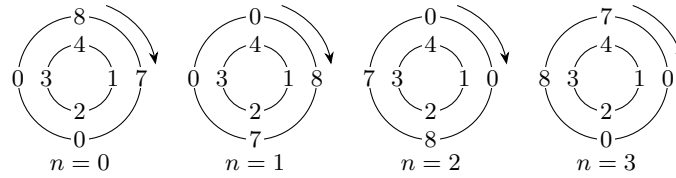
How to read the formula. Fix n , let k take every value from 0 to $N-1$, calculate the wrapped index $(n-k) \bmod N$, multiply the corresponding two values, then add. A **dot product** is just this multiply-matching-entries-and-add operation.

Zero-padding and the origin. First write the ordinary index order $h[0], h[1], \dots$, then append zeros up to N entries. After that, place the entries around the circle in the chosen direction. Changing the drawing direction does not move the origin or change where the extra zero indices belong. Circular reversal keeps entry 0 first, then reads entries $N-1, N-2, \dots, 1$.

Question 6 — Four-point circular convolution with the ring method

Find the 4-point circular convolution of $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$.

Solution. Pad the second input to $h = \{7, 8, 0, 0\}$. Place x clockwise on the inner ring, starting at the right. Place h anticlockwise on the outer ring. Read aligned products and add; then rotate the outer ring one position clockwise for the next output.



At fixed $k = 0, 1, 2, 3$ (right, down, left, up), the outer values and dot products are:

n	$(h[n], h[n-1], h[n-2], h[n-3])$	Dot product with $(1, 2, 3, 4)$
0	$(7, 0, 0, 8)$	$1(7) + 2(0) + 3(0) + 4(8) = 39$
1	$(8, 7, 0, 0)$	$1(8) + 2(7) + 3(0) + 4(0) = 22$
2	$(0, 8, 7, 0)$	$1(0) + 2(8) + 3(7) + 4(0) = 37$
3	$(0, 0, 8, 7)$	$1(0) + 2(0) + 3(8) + 4(7) = 52$

All h indices above are modulo 4. Each row is a right rotation of the preceding row.

$$y_4 = \{39, 22, 37, 52\}_{n=0,1,2,3}.$$

Check: $39 + 22 + 37 + 52 = 150 = (1 + 2 + 3 + 4)(7 + 8)$. The same result follows from $y_4[n] = 7x[n] + 8x[(n-1) \bmod 4]$.

Question 7 — Three-point circular convolution with a negative sample

Find the 3-point circular convolution of $x = \{1, 2, 0\}$ and $h = \{1, -1, 0\}$.

Solution. Circularly reverse h to get $(h[0], h[2], h[1]) = (1, 0, -1)$, then rotate right for each successive output:

n	Rotated h row	Dot product with $(1, 2, 0)$
0	$(1, 0, -1)$	$1(1) + 2(0) + 0(-1) = 1$
1	$(-1, 1, 0)$	$1(-1) + 2(1) + 0(0) = 1$
2	$(0, -1, 1)$	$1(0) + 2(-1) + 0(1) = -2$

$$y_3 = \{1, 1, -2\}_{n=0,1,2}.$$

Check: $1 + 1 - 2 = 0 = (1 + 2 + 0)(1 - 1 + 0)$.

5 Linear versus circular: choose the right length

	Linear convolution	N -point circular convolution
Index behaviour	Finite inputs are zero outside their ranges.	Input indices wrap modulo N .
Output length	$L + M - 1$ listed positions.	Exactly N stored positions.
Method	Flip, shift, multiply, add.	Reverse on a ring, rotate, multiply, add.
Tail	Remains at its own indices.	Folds onto indices with the same remainder modulo N .

Zero-padding means adding zero-valued entries to the end of the listed index range. It leaves the original sample values and their indices unchanged. **Wrapping** means adding contributions whose output indices differ by a multiple of N . When distinct linear-output contributions fold together, this is called **time aliasing** in circular convolution.

Use the stated N and pad finite inputs with zeros as needed. If no N is given, $N = \max(L, M)$ is a common convention; state it explicitly. To reproduce the full linear result by circular convolution, the standard sufficient padding rule is

$$N \geq L + M - 1.$$

Repeating a sequence changes its samples; it is not zero-padding. An LCM may give a common period for periodic sequences, but is not the general padding rule for finite-list convolution.

Question 8 — Linear output, circular wrapping and zero-padding

(a) Calculate and compare the outputs. For $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$, find the linear convolution and obtain the 4-point and 5-point circular results from it.

Solution. The linear output has $4 + 2 - 1 = 5$ positions. Calculate every sample:

$$\begin{aligned} y_{\text{lin}}[0] &= 1(7) = 7, \\ y_{\text{lin}}[1] &= 1(8) + 2(7) = 22, \\ y_{\text{lin}}[2] &= 2(8) + 3(7) = 37, \\ y_{\text{lin}}[3] &= 3(8) + 4(7) = 52, \\ y_{\text{lin}}[4] &= 4(8) = 32. \end{aligned}$$

Hence $y_{\text{lin}} = \{7, 22, 37, 52, 32\}$ at $n = 0, \dots, 4$. For $N = 4$, index 4 wraps to index 0, so

$$y_4[0] = 7 + 32 = 39, \quad y_4 = \{39, 22, 37, 52\}.$$

For $N = 5$, pad both inputs to length 5. No nonzero tail wraps into the start, giving

$$y_5 = \{7, 22, 37, 52, 32\} = y_{\text{lin}}.$$

Check: both totals are $150 = (1 + 2 + 3 + 4)(7 + 8)$. The minimum length preserving all five linear positions is $N = 5$.

(b) Selecting a length from the input sizes. Two finite lists have lengths $L = 4$ and $M = 6$. Find $\max(L, M)$, $\text{lcm}(L, M)$ and the minimum length guaranteed to reproduce their full linear convolution using circular convolution.

Solution. The larger input length is $\max(4, 6) = 6$. Since $4 = 2^2$ and $6 = 2 \cdot 3$, their least common multiple is $2^2 \cdot 3 = 12$. The linear output length is

$$N_{\text{min}} = L + M - 1 = 4 + 6 - 1 = 9.$$

At $N = 6$, possible linear samples at indices 6, 7, 8 fold onto 0, 1, 2, respectively. At $N = 9$, append five zeros to the length-4 input and three zeros to the length-6 input. All nine output positions remain distinct. At $N = 12$, the linear output is also reproduced, followed by three zeros, but the extra padding is unnecessary.

6 Z-transform: attach a power to each sample

A **transform** describes the same signal in a different mathematical form. Instead of listing sample values, the Z-transform packages them into an expression in a variable z . Lowercase $x[n]$ names the sequence; uppercase $X(z)$ names its transform. A **coefficient** is the number multiplying a power, so in $3z^{-2}$ the coefficient is 3 and the exponent is -2 .

The introductory Z-transform represents each sample as a coefficient and its index as a power:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

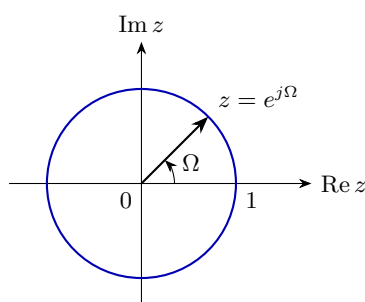
Write the indices first, form $x[n]z^{-n}$ for every nonzero sample, and add. Negative indices give positive powers of z . For an infinite series, state its convergence condition. Only this basic convergence condition is used here.

What is z ? Reading the circle in the assignment

The transform variable z is a complex number. In polar form,

$$z = re^{j\Omega} = r(\cos \Omega + j \sin \Omega), \quad j^2 = -1.$$

Here $r = |z|$ is the distance from the origin and Ω is the angle in radians. The **unit circle** has radius 1, so $z = e^{j\Omega}$ on that circle. The sketch describes the transform variable, not a time-domain signal.



For each sample, the exponent is *minus its index*: $n = -1$ gives z^1 , $n = 0$ gives $z^0 = 1$, and $n = 1$ gives $z^{-1} = 1/z$. The **region of convergence** (ROC) is the set of z values where the transform converges. The finite sequences below need only exclude $z = 0$ (among finite complex values), because their expressions contain negative powers.

Question 9 — Finite-sequence Z-transforms and convolution

(a) Assignment Z-transform Q1: $x = \{2, 4, 7\}_{n=0,1,2}$. Find $X(z)$ directly from the definition.

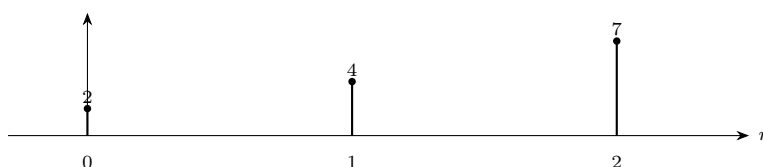
Solution. The first value is at $n = 0$, so

n	0	1	2
$x[n]$	2	4	7
$x[n]z^{-n}$	$2z^0$	$4z^{-1}$	$7z^{-2}$

Adding all three terms,

$$\begin{aligned} X(z) &= \sum_{n=0}^2 x[n]z^{-n} = 2 + 4z^{-1} + 7z^{-2} \\ &= 2 + \frac{4}{z} + \frac{7}{z^2} = \boxed{\frac{2z^2 + 4z + 7}{z^2}}, \quad z \neq 0. \end{aligned}$$

Assignment sequence (a)



Check: $X(1) = 2 + 4 + 7 = 13$, equal to the sum of the samples.

(b) Assignment Z-transform Q2: $x = \{3, 4, 7\}_{n=-1,0,1}$. The origin arrow is under the middle value 4. Find $X(z)$.

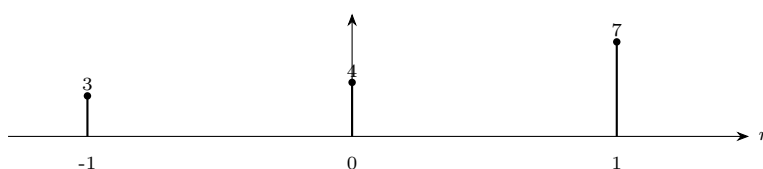
Solution. The indices are now $-1, 0, 1$, not $0, 1, 2$:

n	-1	0	1
$x[n]$	3	4	7
$x[n]z^{-n}$	$3z$	4	$7z^{-1}$

Therefore

$$\begin{aligned} X(z) &= \sum_{n=-1}^1 x[n]z^{-n} = 3z + 4 + 7z^{-1} \\ &= 3z + 4 + \frac{7}{z} = \boxed{\frac{3z^2 + 4z + 7}{z}}, \quad z \neq 0. \end{aligned}$$

Assignment sequence (b)



Check: $X(1) = 3 + 4 + 7 = 14$. The sample at negative index produces a *positive* power of z ; moving the origin changes the transform.

Question 9 continued: (c) Convolution through Z-transforms

Find $X(z)$ and $H(z)$ for

$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1},$$

and multiply them to obtain the transform of their linear convolution.

Solution. Attach z^{-n} to each sample:

Index n	-2	-1	0	1
$x[n]$	1	2	2	1
$x[n]z^{-n}$	z^2	$2z$	2	z^{-1}

Therefore

$$\boxed{X(z) = z^2 + 2z + 2 + z^{-1}}, \quad \boxed{H(z) = z + 3 + 2z^{-1}}.$$

When multiplying transforms, exponents add. The coefficient at z^{-n} is $\sum_k x[k]h[n-k]$, so convolution corresponds to multiplication: $Y(z) = X(z)H(z)$.

$$\begin{aligned} Y(z) &= z^2(z + 3 + 2z^{-1}) + 2z(z + 3 + 2z^{-1}) \\ &\quad + 2(z + 3 + 2z^{-1}) + z^{-1}(z + 3 + 2z^{-1}) \\ &= (z^3 + 3z^2 + 2z) + (2z^2 + 6z + 4) \\ &\quad + (2z + 6 + 4z^{-1}) + (1 + 3z^{-1} + 2z^{-2}) \\ &= \boxed{z^3 + 5z^2 + 10z + 11 + 7z^{-1} + 2z^{-2}}. \end{aligned}$$

The coefficients at indices $-3, -2, -1, 0, 1, 2$ give $\{1, 5, 10, 11, 7, 2\}$, agreeing with Question 4. Each expression is finite for finite $z \neq 0$; $z = 0$ is excluded because negative powers occur.

Question 9 continued: (d) Transform of a right-sided exponential

Find the Z-transform of $x[n] = (1/3)^n u[n]$ directly from the definition.

Solution. Since $u[n] = 0$ for $n < 0$ and $u[n] = 1$ for $n \geq 0$,

$$X(z) = \sum_{n=0}^{\infty} (1/3)^n z^{-n} = 1 + \frac{1}{3}z^{-1} + \frac{1}{9}z^{-2} + \frac{1}{27}z^{-3} + \dots$$

This is a geometric series with ratio $r = \frac{1}{3}z^{-1}$. Use $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ when $|r| < 1$:

$$\boxed{X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} = \frac{z}{z - \frac{1}{3}}, \quad |z| > \frac{1}{3}}.$$

Check: the coefficients of $1, z^{-1}, z^{-2}, z^{-3}$ are $1, 1/3, 1/9, 1/27$, exactly the first four samples. More generally, $a^n u[n]$ has transform $1/(1 - az^{-1})$ for $|z| > |a|$, with $a \neq 0$.

7 Continuous-time convolution: the convolution integral

What changes when time is continuous?

A continuous-time signal $x(t)$ has a value at every real time t . In discrete time, we add individual sample products. In continuous time, contributions occur throughout intervals, so we use an **integral**: a continuous accumulation of contributions. For ordinary real-valued functions, a definite integral is the signed area under a curve between two limits.

The continuous-time convolution of $x(t)$ and $h(t)$ is

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

This is the **convolution integral**. In an LTI system, $x(t)$ is the input, $h(t)$ is the impulse response and $y(t)$ is the output. The asterisk $*$ means convolution, not ordinary multiplication and not a time variable.

Symbol	What it means
t	The output time whose value $y(t)$ we want. Hold it fixed while integrating.
τ	The integration variable (Greek letter “tau”). It moves through the input times. The handwritten notes use k for this same role.
$x(\tau)$	The input expressed using that integration variable.
$h(t - \tau)$	The second signal reversed in time and then shifted by t . Replace <i>every</i> occurrence of its argument, including the argument of a step.
$d\tau$	Integrate with respect to τ , not with respect to t .
$-\infty, \infty$	Initially consider every input time; the actual nonzero overlap determines the useful limits.

Flip, shift, multiply, integrate

1. Rewrite the first signal as $x(\tau)$ and keep it fixed.
2. Reverse $h(\tau)$ to get $h(-\tau)$.
3. Shift the reversed signal by t to obtain $h(t - \tau)$.
4. Find the interval where both factors are nonzero.
5. Multiply the two expressions and integrate over that interval. Repeat the reasoning for each range of t where the overlap changes.

The area is under the *product* $x(\tau)h(t - \tau)$, not generally the geometric area between the two input curves. For unit-height rectangular signals, that product is 1 on the overlap, so the integral equals the overlap width.

How unit steps determine the limits

The step $u(\tau)$ is zero before $\tau = 0$ and one afterwards. It acts as a switch multiplying the rest of the expression. Similarly, $u(t - \tau)$ allows only points where its argument is nonnegative:

$$u(\tau) : \tau \geq 0, \quad u(t - \tau) : t - \tau \geq 0 \implies \tau \leq t.$$

Both switches must be on at the same time, so their common interval is $0 \leq \tau \leq t$, which exists only for $t \geq 0$. For $t < 0$, there is no overlap and the output is zero. Setting a step’s argument equal to zero finds its boundary; the *inequality* tells you which side is allowed.

Question 10 — Convolution of two decaying exponentials

From Assignment 2 in the supplied handwritten pages. Given

$$x(t) = e^{-t}u(t), \quad h(t) = e^{-2t}u(t),$$

find $y(t) = x(t) * h(t)$ for all real t .

Solution. Step 1: understand the signals. The number $e \approx 2.71828$ is the base of the natural exponential. For positive t , $e^{-t} = 1/e^t$ decreases towards zero. Both signals start at height 1 at $t = 0$ and are zero for $t < 0$ because of $u(t)$. The second signal decays faster because its exponent is $-2t$.

Step 2: substitute the correct arguments. Replace t with τ in x , and with $t - \tau$ in *both factors* of h :

$$x(\tau) = e^{-\tau}u(\tau), \quad h(t - \tau) = e^{-2(t-\tau)}u(t - \tau).$$

Therefore

$$y(t) = \int_{-\infty}^{\infty} e^{-\tau}u(\tau)e^{-2(t-\tau)}u(t - \tau) d\tau.$$

Step 3: find the common interval. From the two steps, $\tau \geq 0$ and $\tau \leq t$. Hence $y(t) = 0$ for $t < 0$. For $t \geq 0$, the step factors equal 1 on the integration interval, so

$$y(t) = \int_0^t e^{-\tau}e^{-2(t-\tau)} d\tau.$$

Step 4: simplify the exponent carefully. Use $e^ae^b = e^{a+b}$ and expand the brackets:

$$-\tau - 2(t - \tau) = -\tau - 2t + 2\tau = -2t + \tau.$$

Thus the product is $e^{-2t}e^{\tau}$, **not** $e^{-2t}e^{-\tau}$. This corrects the intermediate sign slip in the handwritten calculation. Since t is fixed while integrating over τ , e^{-2t} is a constant and can be taken outside:

$$y(t) = e^{-2t} \int_0^t e^{\tau} d\tau.$$

Step 5: integrate and substitute the limits. An antiderivative of e^{τ} is e^{τ} , because its derivative is itself. The notation $[F(\tau)]_0^t$ means $F(t) - F(0)$:

$$\begin{aligned} y(t) &= e^{-2t}[e^{\tau}]_0^t = e^{-2t}(e^t - e^0) \\ &= e^{-2t}(e^t - 1) = e^{-t} - e^{-2t}, \quad t \geq 0. \end{aligned}$$

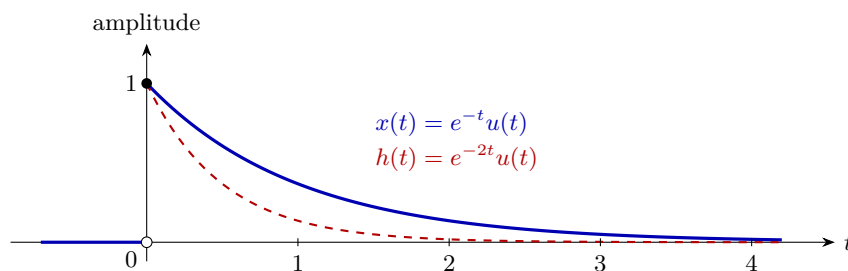
Step 6: state the answer for all time.

$$y(t) = \begin{cases} 0, & t < 0, \\ e^{-t} - e^{-2t}, & t \geq 0, \end{cases} \quad \text{or} \quad y(t) = (e^{-t} - e^{-2t})u(t).$$

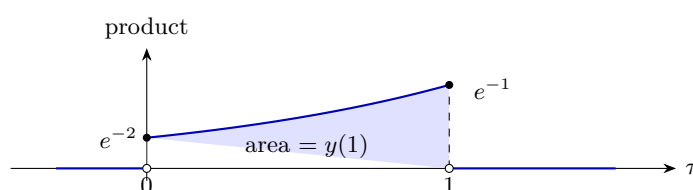
Checks: $y(0) = 1 - 1 = 0$ because the overlap interval has zero width. For $t > 0$, $e^{-t} > e^{-2t}$, so the output is positive. As $t \rightarrow \infty$, both exponentials approach zero, so the output also approaches zero.

Question 10 continued: graphs and interpretation

The two inputs. Both are zero for negative time, jump to 1 at the origin and decay. Solid blue shows $x(t)$; dashed red shows $h(t)$.

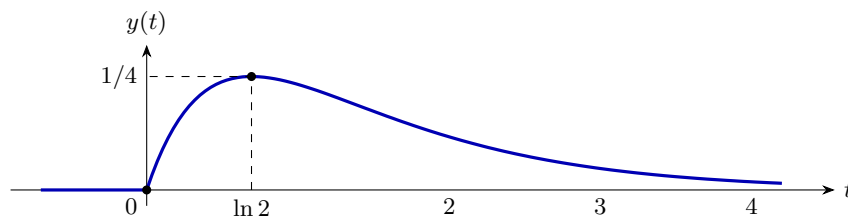


Overlap at the fixed output time $t = 1$. The input $x(\tau)$ allows $\tau \geq 0$, while $h(1 - \tau)$ allows $\tau \leq 1$. Their product is $e^{-2+\tau}$ on $[0, 1]$ and zero elsewhere. The shaded area below is $y(1)$.



Numerically, $y(1) = e^{-1} - e^{-2} \approx 0.23254$. The shaded height is a *product*; the area is the output value at the fixed time $t = 1$.

The output over time. A new integral value is obtained for each t , producing the curve below. It starts at zero, rises, then decays; it is not just the pointwise product e^{-3t} .



The marked point has $t = \ln 2 \approx 0.693$ and $y(t) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. Here $\ln$ is the natural logarithm, the inverse of the exponential: $e^{\ln 2} = 2$.

Method to remember: substitute arguments, solve the step inequalities, intersect their allowed intervals, simplify the product, integrate, and include the zero-output time range. If both ordinary signals are zero for negative time, the overlap is contained in $[0, t]$ for $t \geq 0$; additional finite-duration limits can make it narrower.